

Next Week

HW #6 posted, due (Tues 4/8)

EXAM #2 – “ALL CALCULUS” (Wednesday 4/9)

In class, In person

Blank excel, basic calculator, scratch paper

Elasticity of demand equation provided

Chapter 4

Optimization

L'Hôpital's Rule and the Indeterminate Form 0/0

mathematically
evaluating a limit

$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ is a 0/0 indeterminate form but it can be evaluated using algebraic simplification.

$$\frac{2^2 - 4}{2 - 2} = \frac{0}{0} = \frac{(x-2)(x+2)}{(x-2)} = \lim_{x \rightarrow 2} (x+2) = 2+2=4$$

$$\lim_{x \rightarrow 1} \frac{e^x - e}{x - 1} = \frac{0}{0}$$

L'Hôpital's Rule

$$\lim_{x \rightarrow 1} \frac{e^x - e}{x - 1}$$

Limits such as the one shown can be evaluated using **L'Hôpital's Rule**:

For c a real number,
if $\lim_{x \rightarrow c} f(x) = 0$ and $\lim_{x \rightarrow c} g(x) = 0$, then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

*Reevaluate
ratio of
the derivatives
(not quotient rule)*

provided that the second limit exists or is ∞ or $-\infty$.

Example

$$\lim_{x \rightarrow 1} \frac{e^x - e}{x - 1} = \frac{0}{0}$$

Then to

$$\frac{f'(x)}{g'(x)} = \frac{e^x}{1}$$

$$\lim_{x \rightarrow 1} e^x = e^1 \downarrow e$$

Example

$$\lim_{x \rightarrow 1} \frac{2x^3 - 3x^2 + 1}{x^3 - 3x + 2}$$

$$\frac{6x^2 - 6x}{3x^2 - 3}$$

$$\frac{12x - 6}{6x - 3} = \frac{6}{6} = 1$$

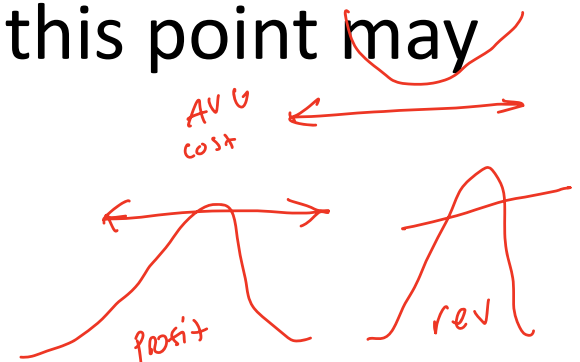
Point of Diminishing Returns

If a company decides to increase spending on advertising, they would expect sales to increase:

At first, sales will increase at an increasing rate and then increase at a decreasing rate. **The value of x where the rate of change of sales changes from increasing to decreasing is called the point of diminishing returns-**

This is also the point where the rate of change has a maximum value. Money spent after this point may increase sales, but at a lower rate...

"Optimize" a "rate"



advertising campaign

A store is selling 200 TVs monthly. If the store invests ***\$x thousand in an advertising campaign***, the company estimates that sales will increase to:

Sales
function:

$$N(x) = 3x^3 - 0.25x^4 + 200$$

$$0 \leq x \leq 9$$

$x \neq \text{production}$

① rate of change of sales
 $N'(x) =$

x

"function"

→ the
derivative

What is the point of diminishing returns and the maximum rate of change of sales?

advertising campaign

total sales

$$N(x) = 3x^3 - 0.25x^4 + 200 \quad 0 \leq x \leq 9$$

The rate of change of sales with respect to advertising expenditures is:

Derivative

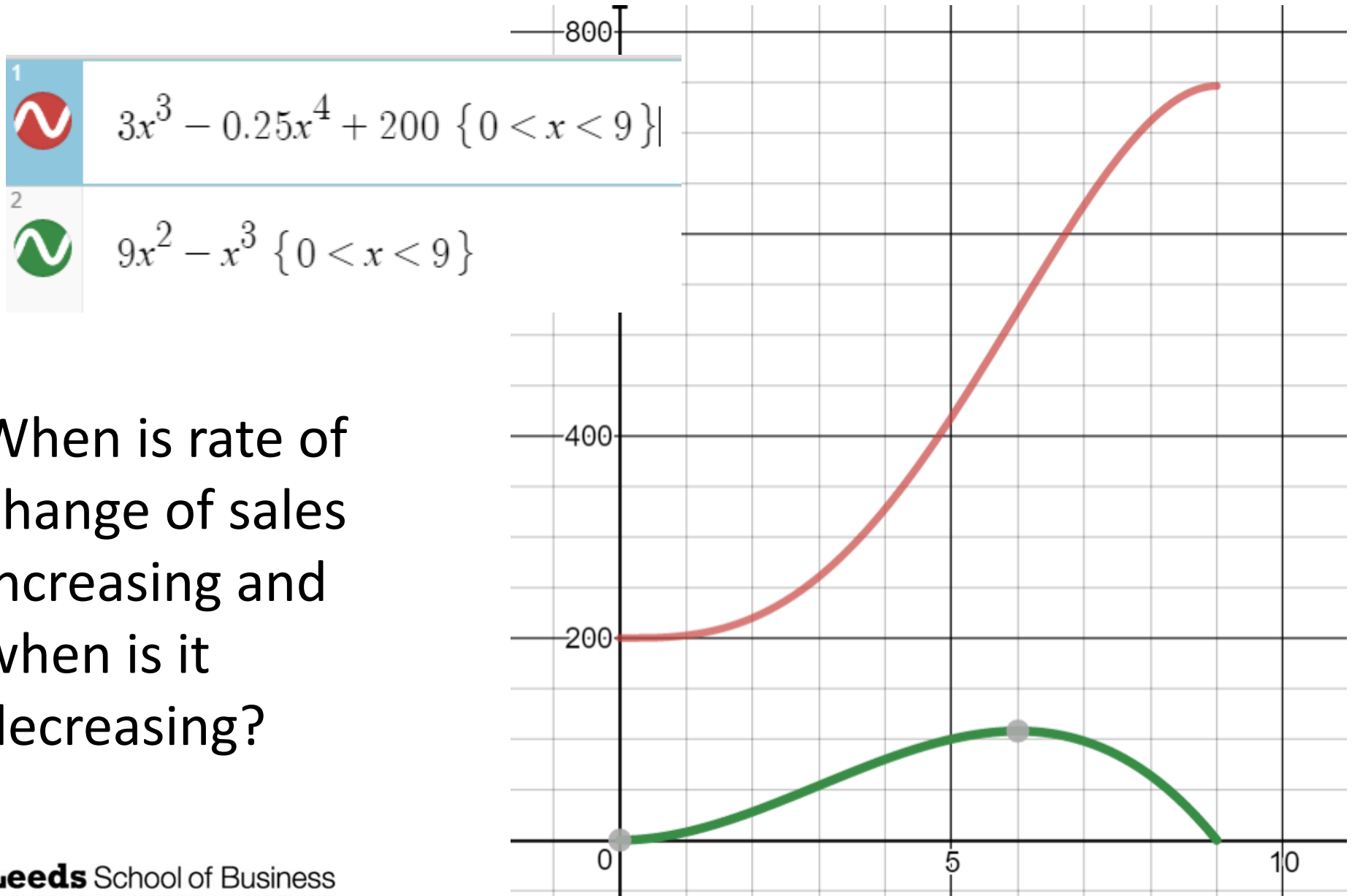
$$N'(x) = 9x^2 - x^3$$

RATE
OF CHANGE
OF SALES

derivative
again

$$18x - 3x^2 = 3x(6-x)$$

advertising campaign



When is rate of change of sales increasing and when is it decreasing?

SOLVER:

Objective: max min

Value of a

A B)

cell = 1

cell f(x) cell b
2 formula:

Set: A1

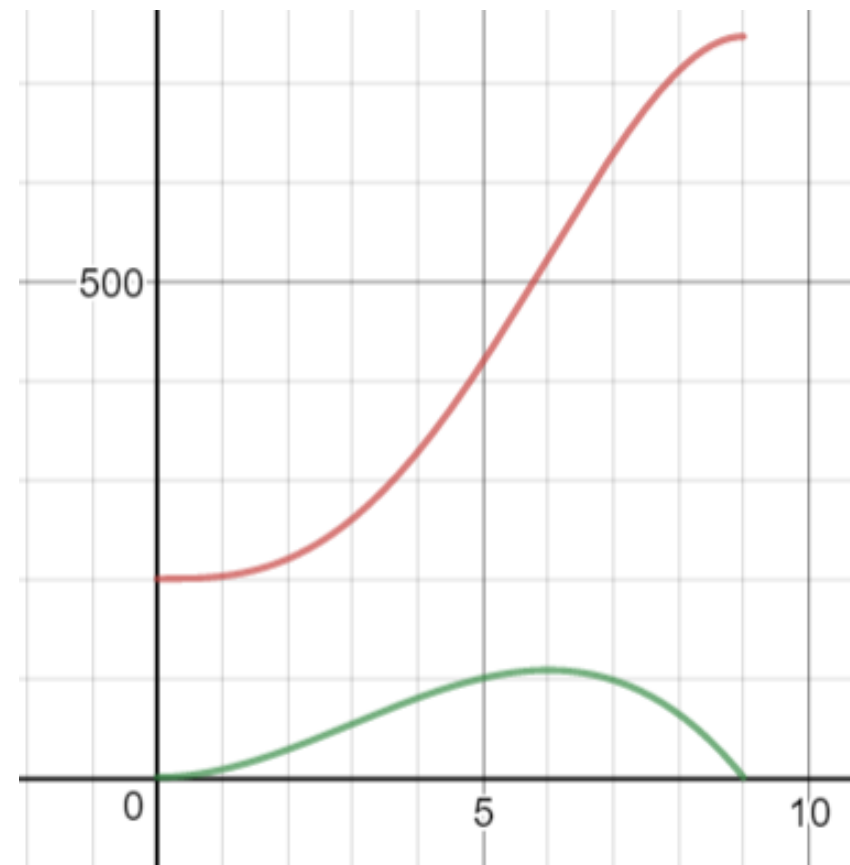
B: change

advertising campaign

What is the point of diminishing returns and the maximum rate of change of sales?

$$N(x) = 3x^3 - 0.25x^4 + 200$$

$$N'(x) = 9x^2 - x^3 = x^2(9 - x)$$



advertising campaign

What is the point of diminishing returns and the maximum rate of change of sales?

$$N'(x) = 9x^2 - x^3$$

max rate of change of sales:

Average Cost & Marginal Cost

marginal
cost = derivative

Total daily cost of producing x mountain bikes is:

TOTAL
COST
FUNCTION

$$C(x) = 1000 + 5x + 0.1x^2$$

AVG COST $\rightarrow \bar{C}(x) = \frac{1000 + 5x + 0.1x^2}{x}$

$\rightarrow$ find minimum avg cost

and the average cost per unit is $C(x) / x$

$$\bar{C}(x) = \frac{1000}{x} + 5 + 0.1x$$

bring up (switch the signs)

= total quantity

$$1000x^{-1} + 0.1 = \frac{1000}{x^2} + 0.1$$

to make pos

find the "minimum" average cost (in dollars)!

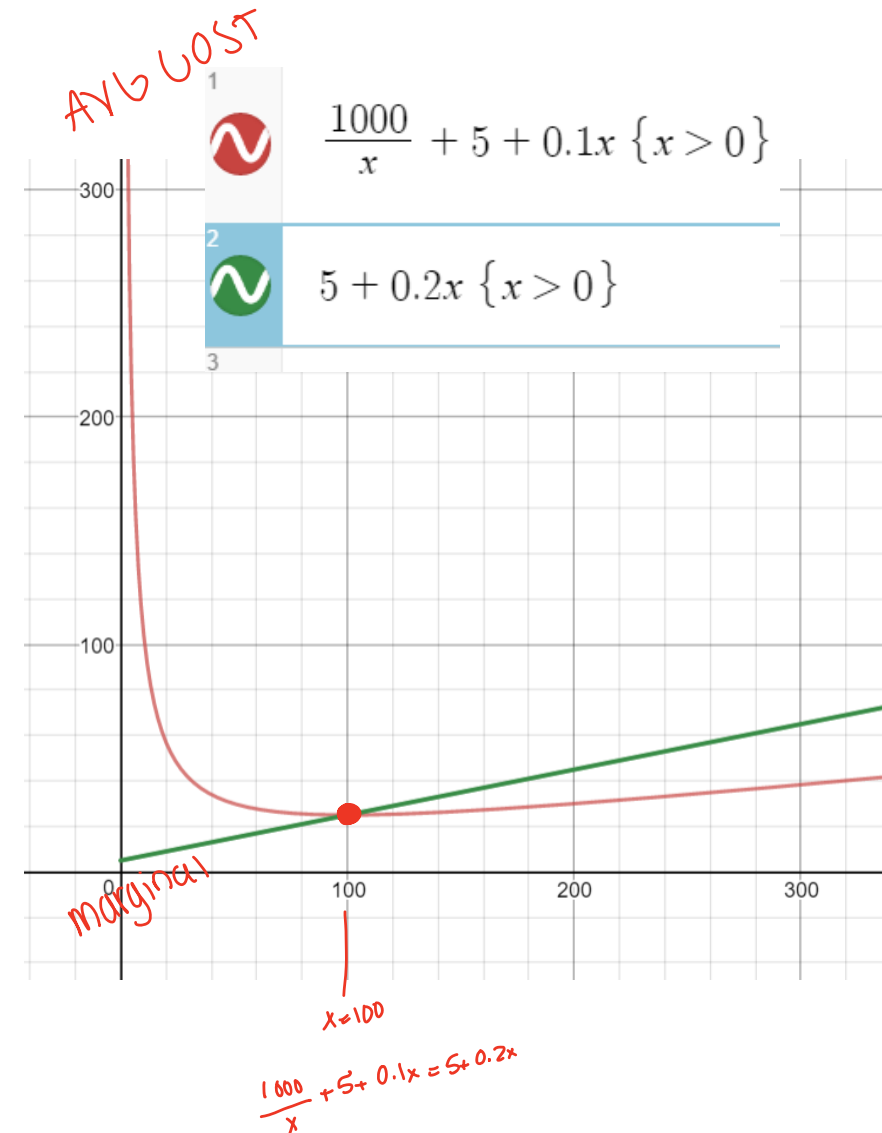
Average Cost & Marginal Cost

CONCEPTUALIZE

$$C(x) = 1000 + 5x + 0.1x^2$$

$$\bar{C}(x) = \frac{1000}{x} + 5 + 0.1x$$

$$C'(x) = 5 + 0.2x$$



Average Cost & Marginal Cost

find the minimum average cost:

The minimum average cost occurs when
average cost = the marginal cost

$$C(x) = 1000 + 5x + 0.1x^2$$

$$\bar{C}(x) = \frac{1000}{x} + 5 + 0.1x$$

Average Cost & Marginal Cost

find the minimum average cost:

The minimum average cost occurs when
average cost = the marginal cost

$$\bar{C}(x) = \frac{1000}{x} + 5 + 0.1x \qquad C'(x) = 5 + 0.2x$$

$$5 + 0.2x = 1000/x + 5 + 0.1x$$

X = 100 bikes

Average Cost & Marginal Cost

find the minimum average cost:

$$\bar{C}(x) = \frac{1000}{x} + 5 + 0.1x$$

$$C(100) = \frac{1000}{100} + 5 + (0.1)(100) = 10 + 5 + 10 = \$25$$

When $x = 100$ bikes, $\bar{C}(100) = 1000/100 + 5 + 0.1(100)$
 $= \$25$

Pizza Cost

The table below gives the total daily costs y (in dollars) of producing x pizzas at various production levels

number of pizzas	total costs
x	y
50	595
100	755
150	1110
200	1380
250	1875
300	2410

Find a quadratic equation for the total cost

Find minimum average cost in \$

Open the file: "Calculus - section 4.4.xls"

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<i>Coefficients</i>	
Intercept	489.5
X Variable 1	1.1725
X Variable 2	0.017392857

$$C(x) = 0.0174x^2 + 1.1725x + 489.5$$

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Data analysis → regression

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$$C(x) = 0.0174x^2 + 1.1725x + 489.5$$
$$= \text{LINEST}(C5:C10, B5:B10^{\{1,2\}})$$

Average Cost

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x	y
50	595
100	755
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$$C(x) = 0.0174x^2 + 1.1725x + 489.5$$

$$\bar{C}(x) = C(x)/x$$

$$\bar{C}(x) = 0.0174x + 1.1725 + 489.5/x$$

Minimum Average Cost

The table below gives the total daily costs y (in dollars) of producing x pizzas at various production levels

$$\bar{C}(x) = C(x)/x$$

$$\bar{C}(x) = 0.0174x + 1.1725 + 489.5/x$$

$$\bar{C}'(x) = 0.0174 - 489.5x^{-2}$$

$$\bar{C}'(x) = 0 \text{ when } x = 167.7 \text{ pizzas}$$

Minimum Average Cost

The table below gives the total daily costs y (in dollars) of producing x pizzas at various production levels

$$\overline{C}'(x) = 0 \text{ when } x = 167.7 \text{ pizzas}$$

So, when $x = 167.7$ pizzas:

$$\overline{C}(167.7) = 0.0174x + 1.1725 + 489.5/x = \$7.01$$

=seriesum(x , leading exponent, step, coeffs.)

Gas Consumption

Using the data below (see excel), where "year" = "x" and $N(x)$ is a function that represents natural gas consumption, find a cubic regression equation $N(x)$ for the data given:

x	gas consumption
0	12.0
10	21.1
20	19.9
30	18.7
40	21.9
50	24.1

Your equation will be of the form $ax^3 + bx^2 + cx + d$. Using the equation generated, calculate the rate at which consumption is changing when $x = 60$ years.

Gas Consumption

Using the data below (see excel), where "year" = "x" and N(x) is a function that represents natural gas consumption, find a cubic regression equation N(x) for the data given:

x	gas consumption
0	12.0
10	21.1
20	19.9
30	18.7
40	21.9
50	24.1

$$N(x) = 0.00055x^3 - 0.04x^2 + 1.084x + 12.55$$

Find $N'(x)$, then evaluate $N'(60)$

$$=\text{SERIESSUM}(60,2,-1,\text{coeffs}) = 1.71$$

VEGAS !

A 300 room hotel in Las Vegas is filled to capacity every night at \$80 per room. For each additional \$1 increase in rent, 3 fewer rooms are rented.

If each rented room cost \$10 to service per day, how much should the management charge for each room to maximize gross profit? What is the maximum gross profit?

HINT: Let x be the amount over \$80 for a room

VEGAS! Profit function

Let x be the amount over \$80 for a room

What is the general formula?

Number of rooms rented = $N(x) = 300 - 3x$

Profit per room = $R(x) = (80 + x) - 10 = 70 + x$

Total Profit = $P(x) = N(x) R(x) = (300 - 3x)(70 + x)$

$$P(x) = -3x^2 + 90x + 21000$$

VEGAS !



$$P(x) = -3x^2 + 90x + 21000$$

$$P'(x) = -6x + 90$$

when $P'(x) = 0$, **$x = 15$ (dollars above 80)**

The room charge should be $80 + 15 =$ **\$95**

Total profit $P(15) = \$21,675$

Sandwiches

A deli sells 320 sandwiches per day at a price of \$ 4 each.

A market survey shows that for every \$ 0.10 reduction in the price, 20 more sandwiches will be sold.

How much should the deli charge in order to maximize revenue?

Sandwiches

A deli sells 320 sandwiches per day at a price of \$ 4 each.

A market survey shows that for every \$ 0.10 reduction in the price, 20 more sandwiches will be sold.

$$R(x) = (4 - 0.1x)(320 + 20x)$$

$$R(x) = 1280 + 48x - 2x^2$$

How much should the deli charge in order to maximize revenue?

$R'(x) = 0 = 48 - 4x$, so $x = 12$, which is 12 ten cent reductions. Price should be $4 - 1.20 = \$2.80$

Optimization

A company manufactures and sells x television sets/month

The monthly cost and price-demand equations are:

$$C(x) = 60,000 + 60x$$

$$p(x) = 200 - x/50 \quad \text{for } 0 \leq x \leq 6,000$$

Find the production level that will maximize the revenue, the maximum revenue, and the price that the company needs to charge at that level.

Optimization

$$C(x) = 60,000 + 60x$$

$$p(x) = 200 - x/50$$

The monthly revenue is

$$R(x) = x \cdot p(x) = 200x - x^2/50$$

Differentiate and set to zero:

$$R'(x) = 200 - x/25 = 0$$

$$x = 5000 \quad (\text{production that maximizes revenue})$$

$$R(5000) = \$500,000 \quad (\text{maximum revenue})$$

$$p(5000) = \$100 \quad (\text{price at that level})$$

